

Gebze Technical University

Department of Mechanical Engineering

ME 331 System Dynamics and Vibration

Undamped Free Vibration

Single-degree-of-freedom systems

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Today's topics

1. Modeling: Newton, D'Alembert and virtual work
2. Natural frequency and static equilibrium
3. Initial conditions, amplitude and phase
4. Energy exchange and conservation
5. Torsional vibration
6. Rayleigh: spring mass, U-tube and beam models
7. Assumptions and result checks

From a structure to a vibration model



Choose a coordinate.

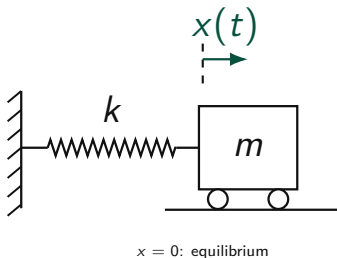
Identify inertia and restoring action.

State the idealization.

3.1 Modeling

Undamped spring-mass model

One coordinate ■ Linear spring ■ Frictionless motion



$$m\ddot{x} + kx = 0$$

Displacement from
equilibrium

3.1 Modeling

D'Alembert and virtual displacement

Newton's law as dynamic equilibrium

$$\sum F_x - m\ddot{x} = 0$$

A compatible imagined displacement at a fixed instant

$$\delta W = (-kx - m\ddot{x}) \delta x = 0$$

Why is δx arbitrary? Which reactions do no work?

3.1 Modeling

Vertical spring–mass system

Measure x from static equilibrium.

$$k\delta_{\text{st}} = mg$$

3.2 Natural frequency

Natural frequency and period

Natural angular frequency

$$\omega_n = \sqrt{\frac{k}{m}}$$

rad/s

Frequency and period

$$f_n = \frac{\omega_n}{2\pi} \quad \text{Hz}$$

$$T_n = \frac{2\pi}{\omega_n} \quad \text{s}$$

How do mass and stiffness change the motion?

3.2 Natural frequency

Frequency from static deflection

Static test ■ Units ■ Parameter effects

$$\omega_n = \sqrt{\frac{k}{m}}, \quad k\delta_{\text{st}} = mg$$

3.3 Initial conditions and response

Response to initial conditions

Two initial conditions determine the motion.

$$x(0) = x_0, \quad \dot{x}(0) = v_0$$

$$x(t) = A_1 \cos(\omega_n t) + A_2 \sin(\omega_n t)$$

3.3 Initial conditions and response

Apply the initial conditions

Find A_1, A_2 ■ Write $x(t)$ ■ Check $t = 0$

3.3 Initial conditions and response

Amplitude and phase

Use the book's cosine form and check the initial conditions.

$$x(t) = A \cos(\omega_n t - \phi)$$

$$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_n}\right)^2}$$

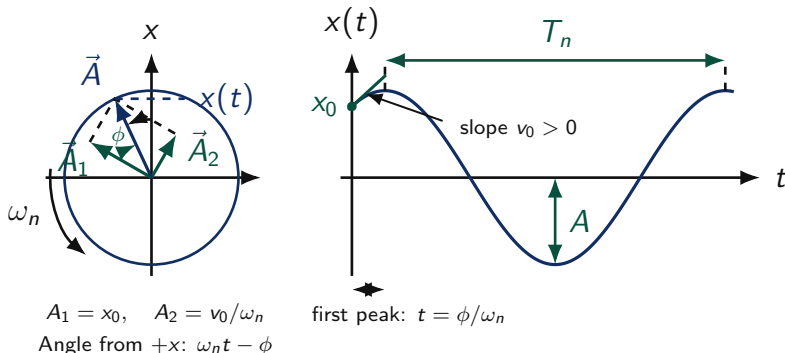
$$\tan \phi = \frac{v_0}{\omega_n x_0}$$

Choose the quadrant from x_0 and v_0 .

3.3 Initial conditions and response

Rotating amplitude vectors

Vertical projection gives $x(t) = A \cos(\omega_n t - \phi)$.



3.4 Energy

Energy exchange

Energy measured relative to static equilibrium

Kinetic energy

$$T = \frac{1}{2}m\dot{x}^2$$

At maximum displacement?

Potential energy

$$U = \frac{1}{2}kx^2$$

At equilibrium?

3.4 Energy

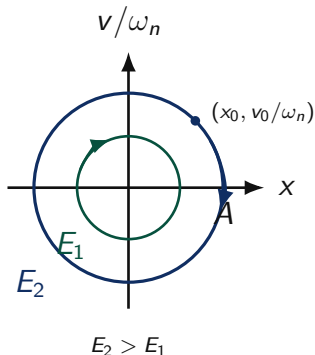
Conservation of energy

Compare two instants of the same motion.

3.4 Energy

Phase plane and energy

Each orbit has constant $E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$.



$$x^2 + \left(\frac{v}{\omega_n}\right)^2 = A^2$$

$$E = \frac{1}{2}kA^2$$

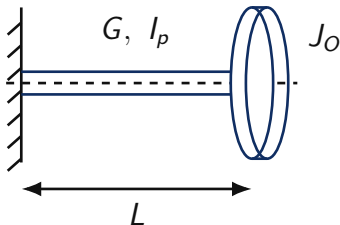
Clockwise motion

Larger orbit: more energy

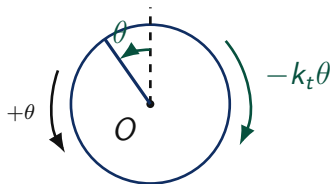
3.5 Torsional vibration

Undamped torsional vibration

Elastic shaft ■ Rigid disk ■ Fixed rotation axis



Shaft inertia neglected



End view: restoring torque

$$k_t = \frac{GI_p}{L}, \quad J_O \ddot{\theta} + k_t \theta = 0$$

3.5 Torsional vibration

Torsional response

Inertia axis ■ Restoring moment ■ Initial conditions

3.6 Rayleigh's method

Rayleigh energy method

Undamped motion about a stable equilibrium

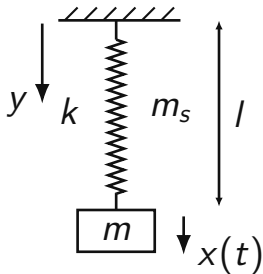
$$T_{\max} = U_{\max}$$

Choose a coordinate. Express every motion using it.

3.6 Rayleigh's method

Including the spring mass

Uniform spring mass m_s ■ Fixed upper end



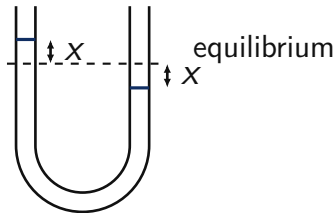
Assume a linear displacement shape

$$u(y, t) = \frac{y}{l} x(t)$$

3.6 Rayleigh's method

U-tube: natural frequency

Uniform area A ■ Liquid density ρ ■ Total length l



$$m_{\text{eq}} = \rho A l$$

$$k_{\text{eq}} = 2\rho g A$$

3.6 Rayleigh's method

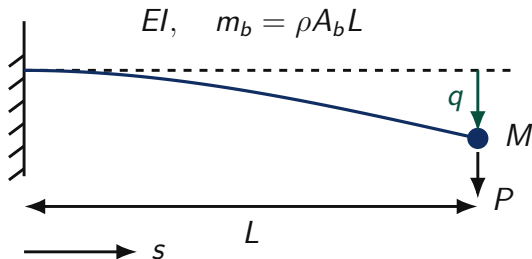
U-tube: energy calculation

Write $T_{\max}$ and $U_{\max}$, then find ω_n .

3.6 Rayleigh's method

Beam model: equivalent stiffness

Uniform cantilever ■ Tip displacement $q(t)$



$$q_{\text{st}} = \frac{PL^3}{3EI} \quad \Rightarrow \quad k_{\text{eq}} = \frac{3EI}{L^3}$$

3.6 Rayleigh's method

Beam model: effective mass

Use the static tip-load shape; normalize it at the tip.

$$w(s, t) = \psi(s)q(t), \quad \psi(s) = \frac{s^2(3L - s)}{2L^3}$$

$$m_{\text{eq}} = M + \int_0^L \rho A_b \psi^2(s) ds$$

$$k_{\text{eq}} = \int_0^L EI [\psi''(s)]^2 ds$$

Same coordinate and shape for both energies. Find ω_R .

3.7 Assumptions and result checks

Model and result checks

Coordinate ■ Units ■ Initial conditions

Whiteboard notes

Whiteboard notes

Whiteboard notes

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